

We have examined the usual representations for both program and data inside the computer. Let us now examine the possible external representations.

EXTERNAL REPRESENTATION OF INFORMATION

The external representation refers to the way information is presented to the *user*, i.e. generally to the programmer. Information may be presented externally in essentially three formats: binary, octal or hexadecimal and symbolic.

1. Binary

It has been seen that information is stored internally in *bytes*, which are sequences of eight *bits* (0's or 1's). It is sometimes desirable to display this internal information directly in its binary format and this is called *binary representation*. One simple example is provided by Light Emitting Diodes (LEDs) which are essentially miniature lights, on the front panel of the microcomputer. In the case of an eight-bit microprocessor, a front panel will typically be equipped with eight LEDs to display the contents of any internal register. (A register is used to hold eight bits of information and will be described in Chapter 2). A lighted LED indicates a one. A zero is indicated by an LED which is not lighted. Such a binary representation may be used for the fine debugging of a complex program, especially if it involves input/output, but is naturally impractical at the human level. This is because in most cases, one likes to look at information in symbolic form. Thus "9" is much easier to understand or remember than "1001". More convenient representations have been devised, which improve the person-machine interface.

2. Octal and Hexadecimal

"Octal" and "hexadecimal" encode respectively three and four binary bits into a unique symbol. In the octal system, any combination of three binary bits is represented by a number between 0 and 7.

"Octal" is a format using three bits, where each combination of three bits is represented by a symbol between 0 and 7: